

Additional Exercise

Table of Contents

1. Introduction

2. Problem Statement

3. Output

1. Introduction

This assignment will help you to consolidate the concepts learnt in the session.

2. Problem Statement

Data Source - http://www.stat.ufl.edu/~winner/datasets.html

Data Set - http://www.stat.ufl.edu/~winner/data/airq402.dat

Data Description - http://www.stat.ufl.edu/~winner/data/airq402.txt

Dataset: airq402.dat

Source: U.S. Department of Transportation

Description: Airfares and passengers for U.S. Domestic Routes for 4th Quarter of 2002.

Variables/Columns

City1 1-3
City2 5-7
Average Fare 11-17
Distance 20-23
Average weekly passengers 26-33
market leading airline 36-37
market share 40-45
Average fare 48-54
Low price airline 57-58
market share 61-66
price 69-75

Assignment Expectations/Steps -

Part 1:

- Load the data using Python Pandas library.
- Do Exploratory Data Analysis of the data and state your insights.

Part 2:

- Treat “Average Fare” – 3rd Column as your Dependent Variable and Rest of the columns as Independent Variable.
- Create Scatter Plot of Independent Variable vs Dependent Variable.
- Based on Scatter Plot see if there is any transformation required for Independent Variable.
- Build Multiple Linear Regression model.
- Get the accuracy score on train and test data.

Part 3:

- Find the most important features of this dataset to predict the average fair.
- Figure out what other model can be applied to improve the model performance.
- What are other ways this data can be improved which will help in improving the prediction
- State your final observations after this complete analysis.

NOTE: The solution shared through Github should contain the source code used and the screenshot of the output.

3. Output

N/A